



Mean Value Theorem

Calculus Worksheet · Grade 11-12

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Learning Objectives

- Verify the hypotheses of the Mean Value Theorem on a closed interval
- Apply the Mean Value Theorem formula to locate the value c
- Interpret c as the point where the tangent line is parallel to the secant line

For each function and interval, verify the Mean Value Theorem applies, then find all values of c that satisfy the conclusion.

1. Find the value of c guaranteed by the Mean Value Theorem for $f(x)=x^2$ on $[0,2]$.

$$f(x) = x^2, [0, 2]$$

Answer: _____

2. Find the value of c guaranteed by the Mean Value Theorem for $f(x)=x^2+2x$ on $[-1,3]$.

$$f(x) = x^2 + 2x, [-1, 3]$$

Answer: _____

3. Find the value of c guaranteed by the Mean Value Theorem for $f(x)=x^3$ on $[0,3]$.

$$f(x) = x^3, [0, 3]$$

Answer: _____

4. Find the value of c guaranteed by the Mean Value Theorem for $f(x)=\sqrt{x}$ on $[1,9]$.

$$f(x) = \sqrt{x}, [1, 9]$$

Answer: _____

5. Find the value of c guaranteed by the Mean Value Theorem for $f(x)=1/x$ on $[1,4]$.

$$f(x) = \frac{1}{x}, [1, 4]$$

Answer: _____

6. Find the value of c guaranteed by the Mean Value Theorem for $f(x)=x^3-x$ on $[0,2]$.

$$f(x) = x^3 - x, [0, 2]$$

Answer: _____

7. Find the value of c guaranteed by the Mean Value Theorem for $f(x)=\ln(x)$ on $[1,e]$.

$$f(x) = \ln(x), [1, e]$$

Answer: _____



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8. Find the value of c guaranteed by the Mean Value Theorem for $f(x)=2x^2-3x+1$ on $[1,4]$.

$$f(x) = 2x^2 - 3x + 1, [1, 4]$$

Answer: _____

9. Find the value of c guaranteed by the Mean Value Theorem for $f(x)=\sin(x)$ on $[0,\pi]$.

$$f(x) = \sin(x), [0, \pi]$$

Answer: _____

10. Find the value of c guaranteed by the Mean Value Theorem for $f(x)=x^2-4x+3$ on $[1,5]$.

$$f(x) = x^2 - 4x + 3, [1, 5]$$

Answer: _____





Remind students that the MVT requires continuity on $[a,b]$ and differentiability on (a,b) before solving for c .

Solutions

1. Find the value of c guaranteed by the Mean Value Theorem for $f(x)=x^2$ on $[0,2]$.

$$f(x) = x^2, [0, 2]$$

- Confirm f is continuous and differentiable everywhere, so MVT applies on the interval.
- Compute $f(b)=4$ and $f(a)=0$, then form the average rate $(4-0)/(2-0)=2$.
- Differentiate to get f prime of x equals $2x$ and set $2c$ equal to 2 .
- Solve to obtain c equals 1 , which lies in the open interval $(0,2)$.

Answer: $c = 1$

2. Find the value of c guaranteed by the Mean Value Theorem for $f(x)=x^2+2x$ on $[-1,3]$.

$$f(x) = x^2 + 2x, [-1, 3]$$

- The polynomial is continuous and differentiable on the interval, so MVT applies.
- Evaluate $f(3)=15$ and $f(-1)=-1$, giving an average rate of $(15-(-1))/(3-(-1))=4$.
- Differentiate to get f prime of x equals $2x+2$ and set $2c+2$ equal to 4 .
- Solve to obtain c equals 1 , which lies in $(-1,3)$.

Answer: $c = 1$

3. Find the value of c guaranteed by the Mean Value Theorem for $f(x)=x^3$ on $[0,3]$.

$$f(x) = x^3, [0, 3]$$

- The cubic is continuous and differentiable on all real numbers.
- Compute $f(3)=27$ and $f(0)=0$, giving an average rate of $27/3=9$.
- Differentiate to get f prime of x equals $3x^2$ and set $3c^2$ equal to 9 .
- Solve $c^2=3$ and take the positive root since c must lie in $(0,3)$, giving c equals the square root of 3 .

Answer: $c = \sqrt{3}$

4. Find the value of c guaranteed by the Mean Value Theorem for $f(x)=\sqrt{x}$ on $[1,9]$.

$$f(x) = \sqrt{x}, [1, 9]$$

- The function is continuous on $[1,9]$ and differentiable on $(1,9)$.
- Compute $f(9)=3$ and $f(1)=1$, giving an average rate of $(3-1)/(9-1)=1/4$.
- Differentiate to get f prime of x equals $1/(2 \text{ square root of } x)$ and set it equal to $1/4$.
- Solve $2 \text{ square root of } c$ equals 4 , so square root of c equals 2 and c equals 4 .

Answer: $c = 4$



5. Find the value of c guaranteed by the Mean Value Theorem for $f(x)=1/x$ on $[1,4]$.

$$f(x) = \frac{1}{x}, [1, 4]$$

- The function is continuous and differentiable on the interval since x is never zero there.
- Compute $f(4)=1/4$ and $f(1)=1$, giving an average rate of $(1/4-1)/(4-1)=-1/4$.
- Differentiate to get f prime of x equals $-1/x^2$ and set $-1/c^2$ equal to $-1/4$.
- Solve $c^2=4$ and take the positive root, giving c equals 2.

Answer: $c = 2$

6. Find the value of c guaranteed by the Mean Value Theorem for $f(x)=x^3-x$ on $[0,2]$.

$$f(x) = x^3 - x, [0, 2]$$

- The polynomial is continuous and differentiable everywhere.
- Compute $f(2)=6$ and $f(0)=0$, giving an average rate of $6/2=3$.
- Differentiate to get f prime of x equals $3x^2-1$ and set $3c^2-1$ equal to 3.
- Solve $c^2=4/3$, so c equals 2 over the square root of 3, which rationalizes to 2 square root of 3 divided by 3.

Answer: $c = \frac{2\sqrt{3}}{3}$

7. Find the value of c guaranteed by the Mean Value Theorem for $f(x)=\ln(x)$ on $[1,e]$.

$$f(x) = \ln(x), [1, e]$$

- The natural log is continuous and differentiable on positive reals.
- Compute $f(e)=1$ and $f(1)=0$, giving an average rate of $(1-0)/(e-1)=1/(e-1)$.
- Differentiate to get f prime of x equals $1/x$ and set $1/c$ equal to $1/(e-1)$.
- Solve to obtain c equals e minus 1, which lies in $(1,e)$.

Answer: $c = e - 1$

8. Find the value of c guaranteed by the Mean Value Theorem for $f(x)=2x^2-3x+1$ on $[1,4]$.

$$f(x) = 2x^2 - 3x + 1, [1, 4]$$

- The polynomial is continuous and differentiable everywhere.
- Compute $f(4)=21$ and $f(1)=0$, giving an average rate of $(21-0)/(4-1)=7$.
- Differentiate to get f prime of x equals $4x-3$ and set $4c-3$ equal to 7.
- Solve to obtain c equals $10/4$ which simplifies to $5/2$.

Answer: $c = \frac{5}{2}$

9. Find the value of c guaranteed by the Mean Value Theorem for $f(x)=\sin(x)$ on $[0,\pi]$.

$$f(x) = \sin(x), [0, \pi]$$

- Sine is continuous and differentiable on all real numbers.
- Compute $f(\pi)=0$ and $f(0)=0$, giving an average rate of $(0-0)/(\pi-0)=0$.
- Differentiate to get f prime of x equals $\cos(x)$ and set $\cos(c)$ equal to 0.
- Solve for c in $(0,\pi)$ to obtain c equals π over 2.

Answer: $c = \frac{\pi}{2}$



10. Find the value of c guaranteed by the Mean Value Theorem for $f(x)=x^2-4x+3$ on $[1,5]$.

$$f(x) = x^2 - 4x + 3, [1, 5]$$

- The polynomial is continuous and differentiable everywhere.
- Compute $f(5)=8$ and $f(1)=0$, giving an average rate of $(8-0)/(5-1)=2$.
- Differentiate to get f prime of x equals $2x-4$ and set $2c-4$ equal to 2.
- Solve to obtain c equals 3, which lies in $(1,5)$.

Answer: $c = 3$

